

Report on "libgamemodeauto.so.0" Preload Errors in Lutris

The error message "ERROR: ld.so: object 'libgamemodeauto.so.0' from LD_PRELOAD cannot be preloaded (wrong ELF class: ELFCLASS64): ignored." indicates a problem with the architecture compatibility of the libgamemodeauto.so.0 library when it is being preloaded, particularly in the context of running games via Lutris on Linux. This report aims to provide an in-depth analysis of the causes, potential solutions, and related issues, especially within the scope of Lutris and Unreal Engine (UE4) games, based on available information and related threads.

Understanding the Error

The error message arises from the dynamic linker (ld.so) failing to preload the libgamemodeauto.so.0 library. This typically occurs because the library's architecture (ELF class) doesn't match the architecture of the process attempting to load it. ELFCLASS64 indicates that the library is compiled for a 64-bit architecture. If a 32-bit process attempts to load a 64-bit library, this error will occur ([Steamcommunity.com](https://steamcommunity.com)).

Gamemode, developed by Feral Interactive, is a set of tools and libraries designed to optimize gaming performance on Linux systems. It allows games to request performance enhancements from the operating system temporarily ([FeralInteractive/gamemode](https://feralinteractive.com/gamemode)). Lutris, a game manager for Linux, often integrates with Gamemode to improve game performance automatically.

Causes and Common Scenarios

Several scenarios can trigger the reported error:

- 1. Architecture Mismatch:** The most common cause is attempting to load a 64-bit version of libgamemodeauto.so.0 into a 32-bit process or vice versa. This mismatch prevents the dynamic linker from correctly loading the library.
- 2. Incorrect Installation:** Gamemode might not be correctly installed or configured for the system's architecture. This can lead to the system trying to use the wrong version of the library.

3. **Lutris Configuration:** Incorrect settings within Lutris can cause it to attempt loading the library in an incompatible manner. For instance, if Lutris is configured to use a 32-bit Wine prefix for a game that requires 64-bit libraries, this error can arise.
4. **Missing Dependencies:** Though less directly related to the ELF class error, missing dependencies can sometimes lead to indirect loading issues. The dynamic linker might fail if dependent libraries are not available.
5. **VirtualGL Configuration:** Similar errors can occur with other libraries preloaded via LD_PRELOAD, such as those used by VirtualGL. A misconfiguration in VirtualGL trying to load incorrect architecture libraries can also cause this error ([NixOS/nixpkgs](#)).

Potential Solutions and Workarounds

Addressing the `libgamemodeauto.so.0` preload error involves verifying architecture compatibility, ensuring correct installation, and adjusting Lutris settings:

1. Verify Architecture:

- Ensure that both the game and the Wine prefix used by Lutris are configured for the correct architecture (32-bit or 64-bit).
- If running a 64-bit system, prioritize using 64-bit Wine prefixes and game installations.

2. Reinstall Gamemode:

- Reinstall Gamemode to ensure that it is correctly configured for your system's architecture. Use the system's package manager to install the appropriate version.
- For Debian-based systems (like MX Linux, as mentioned in one of the source threads), this may involve using `apt`:

```
bash sudo apt update  
sudo apt install gamemode
```

3. Lutris Configuration Adjustments:

- Within Lutris, configure the game to use a Wine prefix that matches the game's architecture requirements.

- Check the Lutris system options to ensure that "Enable Feral Gamemode" is correctly configured. If the option is greyed out, ensure that Gamemode is properly detected by Lutris ([lutris/lutris](https://lutris.net/lutris/)).
- Create or modify the Wine prefix using the Lutris interface to guarantee architecture consistency.

4. Wine Dependencies:

- Ensure all necessary Wine dependencies are installed. Lutris' documentation provides a list of required dependencies for various distributions ([lutris/docs](https://lutris.net/docs/)).
- Install any missing dependencies using the system package manager.

5. Manual LD_PRELOAD Adjustment (Advanced):

- In some cases, it might be necessary to manually adjust the LD_PRELOAD environment variable. However, this should be done with caution.
- Unset or modify the LD_PRELOAD variable before launching the game via Lutris to prevent the problematic library from being preloaded. This can be achieved by editing the game's configuration in Lutris, adding an environment variable: Variable: LD_PRELOAD Value: (Leave Empty or set to a known safe value)

6. GStreamer Issues:

- If GStreamer errors related to wrong ELF class are also present, ensure that the correct GStreamer plugins are installed and compatible with the Wine architecture ([Lutris Forums](https://lutris.net/forums/)).
- Install both 32-bit and 64-bit versions of GStreamer plugins to cover potential compatibility issues.

Lutris and League of Legends

One of the source threads specifically mentions League of Legends. The suggested solution involves:

- 1. Reinstalling League of Legends via Lutris:** Use the Lutris installation script for League of Legends to ensure a correct setup.

2. **Verifying Wine Dependencies:** Confirm that all necessary Wine dependencies are installed as per Lutris' documentation.

This approach aims to provide a clean and consistent environment for League of Legends, reducing the likelihood of architecture mismatches and missing dependencies ([Lutris Forums](#)).

Implications for Unreal Engine (UE4) Games

For UE4 games, the same principles apply:

1. **Wine Prefix:** Ensure the Wine prefix is correctly configured for the target architecture of the UE4 game. Most modern UE4 games are 64-bit, so a 64-bit Wine prefix is generally required.
2. **Dependencies:** UE4 games often have specific dependencies. Use tools like `ldd` on the game executables within the Wine prefix to identify any missing libraries.
3. **Graphics Drivers:** Ensure that the correct graphics drivers are installed for the system. On Linux, this typically involves using proprietary drivers for NVIDIA or AMD cards to maximize performance and compatibility.
4. **Gamemode Integration:** While Gamemode can improve performance, misconfiguration can cause issues. Follow the steps above to ensure it is correctly set up and detected by Lutris.

Practical Example: Fixing the Error on MX Linux

Given that one source specifically mentions MX Linux, here's a step-by-step guide to address the error in that context:

1. **Update the System:** `bash sudo apt update sudo apt upgrade`
2. **Install Gamemode:** `bash sudo apt install gamemode`
3. **Verify Gamemode Installation:** `bash gamemoded -v` This command should return the version number, confirming that Gamemode is installed correctly.
4. **Reconfigure Lutris Wine Prefix:**
 - Open Lutris.
 - Select the game (e.g., League of Legends or a UE4 game).

- Right-click and select "Configure."
- Under "Runner options," ensure that the Wine version is compatible and that the prefix architecture matches the game's requirements (usually 64-bit).
- If necessary, create a new Wine prefix with the correct architecture.

5. Install Wine Dependencies:

- Refer to the Lutris documentation for Wine dependencies.
- Install any missing dependencies using apt.

6. Test the Game:

- Launch the game through Lutris and observe whether the error persists.

Additional Troubleshooting Steps

1. **Check System Logs:** Examine system logs (/var/log/syslog or /var/log/messages) for more detailed error messages that could provide additional clues.
2. **Consult Lutris Forums:** Consult the Lutris forums for specific game-related issues. Other users might have encountered and resolved similar problems.
3. **Use Lutris Debugging Tools:** Lutris has built-in debugging tools that can provide detailed information about the game's environment and potential issues.

Conclusion

The "ERROR: ld.so: object 'libgamemodeauto.so.0' from LD_PRELOAD cannot be preloaded (wrong ELF class: ELFCLASS64): ignored." error typically arises from architecture mismatches or misconfigurations related to Gamemode and Wine within Lutris. Addressing this issue involves verifying architecture compatibility, reinstalling Gamemode, adjusting Lutris settings, and ensuring that all necessary dependencies are installed. By systematically addressing these potential causes, users running Linux distributions, such as MX Linux, can improve their gaming experience and resolve preload errors when launching games via Lutris. Moreover, for Unreal Engine 4 games,

ensuring proper Wine prefix configuration and graphics driver installation is equally vital for optimal performance and to prevent such errors.

References:

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